Apache HTTPD HTTPS Configuration Using a Self-Signed SSL Certificate

1. Objective

Configure Apache HTTPD (`httpd`) to serve HTTPS traffic using a self-signed SSL/TLS certificate.

This procedure covers:

1. Installing the Apache SSL module.
2. Generating a self-signed SSL certificate.
3. Configuring Apache to use the certificate.
4. Validating the Apache configuration.
5. Restarting Apache.
6. Verifying HTTPS access.

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2. Prerequisites

Ensure Apache HTTPD is installed and running.

Check:

```
httpd -v
```

Check the service:

```
sudo systemctl status httpd
```

If Apache is not installed:

```
sudo dnf install httpd -y
```

Start and enable Apache:

```
sudo systemctl enable --now httpd
```

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3. Install the SSL Module

Install `mod_ssl`:

```
sudo dnf install mod_ssl -y
```

Verify that the SSL module is installed:

```
rpm -qa | grep mod_ssl
```

You can also check that Apache recognizes SSL:

```
httpd -M | grep ssl
```

Expected output:

```
ssl_module (shared)
```

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4. Generate a Self-Signed SSL Certificate

Generate a 2048-bit RSA private key and self-signed certificate:

```
sudo openssl req -x509 -nodes -days 365 -newkey rsa:2048 \
-keyout /etc/pki/tls/private/apache-selfsigned.key \
-out /etc/pki/tls/certs/apache-selfsigned.crt
```

During certificate generation, OpenSSL will ask for information such as:

```
Country Name (2 letter code)
State or Province Name
Locality Name
Organization Name
Organizational Unit Name
Common Name
Email Address
```

For the **Common Name (CN)**, enter the hostname or DNS name that will be used to access the Apache server.

For example:

```
Common Name: webserver.example.com
```

For a test server, you may use the server hostname or IP address, but browsers will normally display a certificate warning for a self-signed certificate.

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5. Verify the Certificate and Private Key

Check that the certificate exists:

```
sudo ls -l /etc/pki/tls/certs/apache-selfsigned.crt
```

Check that the private key exists:

```
sudo ls -l /etc/pki/tls/private/apache-selfsigned.key
```

Verify the certificate:

```
sudo openssl x509 \
-in /etc/pki/tls/certs/apache-selfsigned.crt \
-text \
-noout
```

You can also check the certificate validity dates:

```
sudo openssl x509 \
-in /etc/pki/tls/certs/apache-selfsigned.crt \
-noout \
-dates
```

---

6. Set Appropriate Permissions

The private key should be protected because it is sensitive.

Set ownership:

```
sudo chown root:root /etc/pki/tls/private/apache-selfsigned.key
sudo chown root:root /etc/pki/tls/certs/apache-selfsigned.crt
```

Set permissions:

```
sudo chmod 600 /etc/pki/tls/private/apache-selfsigned.key
sudo chmod 644 /etc/pki/tls/certs/apache-selfsigned.crt
```

Verify:

```
sudo ls -l /etc/pki/tls/private/apache-selfsigned.key
sudo ls -l /etc/pki/tls/certs/apache-selfsigned.crt
```

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7. Configure Apache SSL

The SSL configuration is normally located at:

```
/etc/httpd/conf.d/ssl.conf
```

Open the file:

```
sudo vi /etc/httpd/conf.d/ssl.conf
```

Ensure the SSL configuration contains:

```
SSLEngine on

SSLCertificateFile /etc/pki/tls/certs/apache-selfsigned.crt
SSLCertificateKeyFile /etc/pki/tls/private/apache-selfsigned.key
```

A basic SSL virtual host configuration can look like:

```
<VirtualHost *:443>

    ServerName your-server.example.com

    SSLEngine on

    SSLCertificateFile /etc/pki/tls/certs/apache-selfsigned.crt
    SSLCertificateKeyFile /etc/pki/tls/private/apache-selfsigned.key

    DocumentRoot /var/www/html

</VirtualHost>
```

Replace:

```
your-server.example.com
```

with the actual hostname/DNS name of the server.

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8. Check Apache Configuration

Before restarting Apache, always test the configuration:

```
sudo httpd -t
```

Expected result:

```
Syntax OK
```

If you receive an error, **do not restart Apache** until the configuration problem has been corrected.

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9. Restart Apache

After receiving:

```
Syntax OK
```

restart Apache:

```
sudo systemctl restart httpd
```

Check the service:

```
sudo systemctl status httpd
```

You should see:

```
Active: active (running)
```

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10. Verify HTTPS Port

Check whether Apache is listening on TCP port 443:

```
sudo ss -lntp | grep :443
```

Expected output will be similar to:

```
LISTEN 0 511 0.0.0.0:443 0.0.0.0:*
```

You can also check Apache's listening configuration:

```
sudo httpd -S
```

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11. Configure the Firewall

If `firewalld` is running, allow HTTPS traffic:

```
sudo firewall-cmd --permanent --add-service=https
```

Reload the firewall:

```
sudo firewall-cmd --reload
```

Verify:

```
sudo firewall-cmd --list-services
```

You should see:

```
https
```

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12. Test HTTPS Locally

From the Apache server:

```
curl -k https://localhost
```

The `-k` option tells `curl` to accept the self-signed certificate.

You can also test the certificate:

```
openssl s_client -connect localhost:443
```

Exit with: `Ctrl+C`

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13. Test HTTPS From a Client

From another machine, access:

```
https://SERVER_IP
```

or:

```
https://SERVER_HOSTNAME
```

For example:

```
https://10.208.x.x
```

Because this is a **self-signed certificate**, the browser will normally show a warning such as:

```
Your connection is not private
```

or:

```
Warning: Potential Security Risk Ahead
```

This is expected for a self-signed certificate that is not trusted by the client.

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14. Complete Command Sequence

For a quick implementation, the basic command sequence is:

```
# 1. Install SSL module
sudo dnf install mod_ssl -y

# 2. Generate self-signed certificate
sudo openssl req -x509 -nodes -days 365 -newkey rsa:2048 \
-keyout /etc/pki/tls/private/apache-selfsigned.key \
-out /etc/pki/tls/certs/apache-selfsigned.crt

# 3. Check certificate
sudo openssl x509 \
-in /etc/pki/tls/certs/apache-selfsigned.crt \
-noout -dates

# 4. Set permissions
sudo chown root:root /etc/pki/tls/private/apache-selfsigned.key
sudo chmod 600 /etc/pki/tls/private/apache-selfsigned.key

# 5. Edit SSL configuration
sudo vi /etc/httpd/conf.d/ssl.conf

# 6. Test Apache configuration
sudo httpd -t

# 7. Restart Apache
sudo systemctl restart httpd

# 8. Check Apache
sudo systemctl status httpd

# 9. Check HTTPS port
sudo ss -lntp | grep :443

# 10. Test HTTPS
curl -k https://localhost
```

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15. Configuration Summary

Component	Location / Command
Apache configuration	`/etc/httpd/conf.d/ssl.conf`
SSL module	`mod_ssl`
Certificate	`/etc/pki/tls/certs/apache-selfsigned.crt`
Private key	`/etc/pki/tls/private/apache-selfsigned.key`
HTTP port	`80`
HTTPS port	`443`
Configuration test	`sudo httpd -t`
Apache restart	`sudo systemctl restart httpd`
HTTPS test	`curl -k https://localhost`

16. Important Note

A self-signed certificate is suitable for **testing, development, internal lab environments, and temporary HTTPS configuration**.

For a production website or service, use a certificate issued by a trusted Certificate Authority (CA), such as an organization's internal CA or a publicly trusted CA. This avoids browser certificate warnings and provides normal certificate-chain validation.